LV ELECTRICAL

Calculation report

Villa Complex 415V LV Network · Plot 123-456, VILLA, UAE

Doc no.		**Rev**	
Date	15 October 2026		
Prepared		**Checked**	
Approved			

RESULTS SUMMARY

Study	Pass	Warning	Fail
Feeders	36	0	**0**
Earthing	36	0	**0**
Coordination	0	31	**0**

Needs attention (31)

INC-GF → GF-LTG	**WARNING** — Coordination — selectivity limit 4.0 kA, fault 21.8 kA
INC-GF → GF-SKT	**WARNING** — Coordination — selectivity limit 4.0 kA, fault 21.8 kA
INC-GF → GF-HVAC	**WARNING** — Coordination — selectivity limit 4.0 kA, fault 21.8 kA
INC-GF → INC-DBGF1	**WARNING** — Coordination — selectivity limit 4.0 kA, fault 21.8 kA
INC-FF → FF-OFF	**WARNING** — Coordination — selectivity limit 1.6 kA, fault 14.3 kA
INC-FF → FF-IT	**WARNING** — Coordination — selectivity limit 1.6 kA, fault 14.3 kA
INC-FF → FF-LAB	**WARNING** — Coordination — selectivity limit 1.6 kA, fault 14.3 kA
INC-FF → FF-LTG	**WARNING** — Coordination — selectivity limit 1.6 kA, fault 14.3 kA
INC-MCC → MCC-WP	**WARNING** — Coordination — selectivity limit 2.5 kA, fault 19.0 kA
INC-MCC → MCC-FP	**WARNING** — Coordination — selectivity limit 2.5 kA, fault 19.0 kA

…and 21 more — see the tables below.

DESIGN BASIS

Parameter	Value
System	415 V, 3-phase + N, 50 Hz, TN-S (as calculated)
Design ambient temperature	45 °C
Voltage drop limit (source to load)	4 %

Parameter	Value
Cable	Copper; XLPE/PVC/SWA; rated for each circuit's installation method (IEC 60364-5-52)
Standards	IEC 60364-4-41, 4-43, 5-52, 5-54; IEC 60909-based simplified fault levels (preliminary); IEC 60898 / 60947-2 (breakers)
Sizing targets	Transformer ≤ 80% with 20% growth; generator ≤ 80%; PF target 0.95

SYSTEM SUMMARY

Connected load	Maximum demand	Power factor / current	Transformer loading
514 kW	451 kW · 512 kVA	0.88 · 712 A	51 % of 1,000 kVA

Item	Result
Transformer (MDB-1)	Requirement 710 kVA → recommended 1000 kVA; installed 1,000 kVA — adequate
Standby generator	Demand 55 kW / 64 kVA → rating needed 544 kVA (motor start) → recommended 630 kVA / 504 kW (largest motor start: MCC-FP)
Power factor correction (MDB-1)	100 kvar (Central — at the main board (MDB), 25 kvar steps): PF 0.88 → 0.954, 512 → 473 kVA

Board	Voltage (V)	% nominal	Ik″ (kA)	Demand (kW)	Demand (A)	Rating (A)	Loading
MDB-1	415	100.0 %	29.2	451	712	1,600	45 %
SMDB-GF	413	99.5 %	21.8	203	325	630	52 %
DB-GF1	411	99.1 %	10.7	38	59	100	59 %
SMDB-FF	412	99.3 %	14.3	87	135	250	54 %
MCC-1	413	99.4 %	19.0	130	210	400	53 %

FEEDER CALCULATIONS

36 pass · 0 warning · **0 fail**

Circuit	Board	Ib (A)	Breaker	Iz (A)	Ib ≤ In ≤ Iz	Vd total (%)	Vd	Ik″ at breaker (kA)	Icu (kA)	Icu	Status
INC-GF	MDB-1	327	500 A MCCB	525	PASS	0.46	PASS	29.2	36	PASS	PASS
INC-FF	MDB-1	135	200 A MCCB	266	PASS	0.75	PASS	29.2	36	PASS	PASS

Circuit	Board	Ib (A)	Breaker	Iz (A)	Ib ≤ In ≤ Iz	Vd total (%)	Vd	Ik″ at breaker (kA)	Icu (kA)	Icu	Status
INC-MCC	MDB-1	210	315 A MCCB	342	PASS	0.58	PASS	29.2	36	PASS	PASS
DB-EV	MDB-1	44	63 A C	70	PASS	2.38	PASS	29.2	36	PASS	PASS
DB-PV	MDB-1	70	100 A MCCB	120	PASS	0.76	PASS	29.2	36	PASS	PASS
GF-LTG	SMDB-GF	63	80 A MCCB	91	PASS	1.47	PASS	21.8	25	PASS	PASS
GF-SKT	SMDB-GF	27	40 A C	41	PASS	2.37	PASS	21.8	25	PASS	PASS
GF-HVAC	SMDB-GF	177	250 A MCCB	266	PASS	1.16	PASS	21.8	25	PASS	PASS
FF-OFF	SMDB-FF	43	63 A C	70	PASS	1.66	PASS	14.3	25	PASS	PASS
FF-IT	SMDB-FF	29	40 A C	41	PASS	2.04	PASS	14.3	25	PASS	PASS
FF-LAB	SMDB-FF	57	80 A MCCB	91	PASS	1.62	PASS	14.3	25	PASS	PASS
FF-LTG	SMDB-FF	19	25 A C	52	PASS	2.57	PASS	14.3	25	PASS	PASS
MCC-WP	MCC-1	121	160 A MCCB	174	PASS	1.03	PASS	19.0	25	PASS	PASS
MCC-FP	MCC-1	89	125 A MCCB	146	PASS	1.12	PASS	19.0	25	PASS	PASS
INC-DBGF1	SMDB-GF	60	100 A MCCB	146	PASS	0.92	PASS	21.8	25	PASS	PASS
DB-GF1-Y1	DB-GF1	5	6 A C	31	PASS	1.48	PASS	10.7	15	PASS	PASS
DB-GF1-R1	DB-GF1	6	10 A C	31	PASS	1.67	PASS	10.7	15	PASS	PASS
DB-GF1-Y2	DB-GF1	4	6 A C	31	PASS	1.49	PASS	10.7	15	PASS	PASS
DB-GF1-B6	DB-GF1	7	10 A C	41	PASS	1.44	PASS	10.7	15	PASS	PASS
DB-GF1-Y5	DB-GF1	9	16 A C	41	PASS	1.70	PASS	10.7	15	PASS	PASS
DB-GF1-B7	DB-GF1	6	10 A C	41	PASS	1.45	PASS	10.7	15	PASS	PASS
DB-GF1-R3	DB-GF1	28	40 A C	52	PASS	2.79	PASS	10.7	15	PASS	PASS
DB-GF1-B3	DB-GF1	14	20 A C	41	PASS	2.32	PASS	10.7	15	PASS	PASS
DB-GF1-R6	DB-GF1	5	6 A C	41	PASS	1.51	PASS	10.7	15	PASS	PASS
DB-GF1-B4	DB-GF1	12	16 A C	41	PASS	1.79	PASS	10.7	15	PASS	PASS
DB-GF1-Y4	DB-GF1	12	16 A C	41	PASS	1.89	PASS	10.7	15	PASS	PASS

Circuit	Board	Ib (A)	Breaker	Iz (A)	Ib ≤ In ≤ Iz	Vd total (%)	Vd	Ik″ at breaker (kA)	Icu (kA)	Icu	Status
DB-GF1-B5	DB-GF1	12	16 A C	41	PASS	1.99	PASS	10.7	15	PASS	PASS
DB-GF1-R5	DB-GF1	7	10 A C	41	PASS	1.67	PASS	10.7	15	PASS	PASS
DB-GF1-Y8	DB-GF1	1	6 A C	41	PASS	1.05	PASS	10.7	15	PASS	PASS
DB-GF1-Y3	DB-GF1	14	20 A C	41	PASS	2.49	PASS	10.7	15	PASS	PASS
DB-GF1-R4	DB-GF1	12	16 A C	41	PASS	2.18	PASS	10.7	15	PASS	PASS
DB-GF1-B2	DB-GF1	3	6 A C	31	PASS	1.33	PASS	10.7	15	PASS	PASS
DB-GF1-Y7	DB-GF1	5	6 A C	41	PASS	1.35	PASS	10.7	15	PASS	PASS
DB-GF1-B1	DB-GF1	5	6 A C	31	PASS	2.01	PASS	10.7	15	PASS	PASS
DB-GF1-R7	DB-GF1	3	6 A C	41	PASS	1.36	PASS	10.7	15	PASS	PASS
DB-GF1-Y6	DB-GF1	9	16 A C	41	PASS	1.89	PASS	10.7	15	PASS	PASS

EARTHING — FAULT LOOP IMPEDANCE AND DISCONNECTION

36 pass · 0 warning · **0 fail**

Circuit	CPC (mm²)	Zs (Ω)	Max Zs (Ω)	If (A)	Ia (A)	Required (s)	Disconnection	CPC min, each (mm²)	Status
INC-GF	150	0.0144	0.0455	15804	5,000	5	< 0.1 s	34.9	PASS
INC-FF	50	0.0419	0.1138	5433	2,000	5	< 0.1 s	12.0	PASS
INC-MCC	95	0.0204	0.0723	11150	3,150	5	< 0.1 s	24.7	PASS
DB-EV	10	0.2656	0.3613	857	630	0.4	< 0.1 s	1.9	PASS
DB-PV	16	0.0701	0.2276	3246	1,000	5	< 0.1 s	7.2	PASS
GF-LTG	16	0.0932	0.2845	2442	800	5	< 0.1 s	5.4	PASS
GF-SKT	4	0.3969	0.5691	573	400	0.4	< 0.1 s	1.3	PASS
GF-HVAC	50	0.0398	0.0910	5724	2,500	5	< 0.1 s	12.7	PASS
FF-OFF	10	0.1505	0.3613	1512	630	0.4	< 0.1 s	3.3	PASS
FF-IT	4	0.2616	0.5691	870	400	0.4	< 0.1 s	1.9	PASS

Circuit	CPC (mm²)	Zs (Ω)	Max Zs (Ω)	If (A)	Ia (A)	Required (s)	Disconnection	CPC min, each (mm²)	Status
FF-LAB	16	0.1237	0.2845	1840	800	5	< 0.1 s	4.1	PASS
FF-LTG	6	0.2991	0.9105	761	250	0.4	< 0.1 s	1.7	PASS
MCC-WP	25	0.0455	0.1423	4997	1,600	5	< 0.1 s	11.1	PASS
MCC-FP	16	0.0685	0.1821	3325	1,250	5	< 0.1 s	7.4	PASS
INC-DBGF1	16	0.0709	0.2276	3209	1,000	5	< 0.1 s	7.1	PASS
DB-GF1-Y1	2.5	0.3901	455.2407	584	1	0.4	< 0.1 s	1.3	PASS
DB-GF1-R1	2.5	0.4256	455.2407	535	1	0.4	< 0.1 s	1.2	PASS
DB-GF1-Y2	2.5	0.4612	455.2407	494	1	0.4	< 0.1 s	1.1	PASS
DB-GF1-B6	4	0.2693	1517.4690	845	0	0.4	< 0.1 s	1.9	PASS
DB-GF1-Y5	4	0.2914	1517.4690	781	0	0.4	< 0.1 s	1.7	PASS
DB-GF1-B7	4	0.3135	1517.4690	726	0	0.4	< 0.1 s	1.6	PASS
DB-GF1-R3	6	0.2476	1517.4690	919	0	0.4	< 0.1 s	2.0	PASS
DB-GF1-B3	4	0.3356	1517.4690	678	0	0.4	< 0.1 s	1.5	PASS
DB-GF1-R6	4	0.3798	1517.4690	599	0	0.4	< 0.1 s	1.3	PASS
DB-GF1-B4	4	0.2693	1517.4690	845	0	0.4	< 0.1 s	1.9	PASS
DB-GF1-Y4	4	0.2914	1517.4690	781	0	0.4	< 0.1 s	1.7	PASS
DB-GF1-B5	4	0.3135	1517.4690	726	0	0.4	< 0.1 s	1.6	PASS
DB-GF1-R5	4	0.3577	1517.4690	636	0	0.4	< 0.1 s	1.4	PASS
DB-GF1-Y8	4	0.3687	1517.4690	617	0	0.4	< 0.1 s	1.4	PASS
DB-GF1-Y3	4	0.3687	1517.4690	617	0	0.4	< 0.1 s	1.4	PASS
DB-GF1-R4	4	0.3577	1517.4690	636	0	0.4	< 0.1 s	1.4	PASS
DB-GF1-B2	2.5	0.4612	455.2407	494	1	0.4	< 0.1 s	1.1	PASS
DB-GF1-Y7	4	0.3135	1517.4690	726	0	0.4	< 0.1 s	1.6	PASS
DB-GF1-B1	2.5	0.6923	455.2407	329	1	0.4	< 0.1 s	0.7	PASS

Circuit	CPC (mm²)	Zs (Ω)	Max Zs (Ω)	If (A)	Ia (A)	Required (s)	Disconnection	CPC min, each (mm²)	Status
DB-GF1-R7	4	0.4019	1517.4690	566	0	0.4	< 0.1 s	1.3	PASS
DB-GF1-Y6	4	0.3466	1517.4690	657	0	0.4	< 0.1 s	1.5	PASS

PROTECTION COORDINATION

0 pass · 31 warning · **0 fail**

Upstream	Downstream	In ratio	Fault at downstream (kA)	Selectivity limit (kA)	Short circuit	Status
INC-GF (500 A)	GF-LTG (80 A)	6.25	21.8	4.0	Partial	**WARNING**
INC-GF (500 A)	GF-SKT (40 A)	12.50	21.8	4.0	Partial	**WARNING**
INC-GF (500 A)	GF-HVAC (250 A)	2.00	21.8	4.0	Partial	**WARNING**
INC-GF (500 A)	INC-DBGF1 (100 A)	5.00	21.8	4.0	Partial	**WARNING**
INC-FF (200 A)	FF-OFF (63 A)	3.17	14.3	1.6	Partial	**WARNING**
INC-FF (200 A)	FF-IT (40 A)	5.00	14.3	1.6	Partial	**WARNING**
INC-FF (200 A)	FF-LAB (80 A)	2.50	14.3	1.6	Partial	**WARNING**
INC-FF (200 A)	FF-LTG (25 A)	8.00	14.3	1.6	Partial	**WARNING**
INC-MCC (315 A)	MCC-WP (160 A)	1.97	19.0	2.5	Partial	**WARNING**
INC-MCC (315 A)	MCC-FP (125 A)	2.52	19.0	2.5	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-Y1 (6 A)	16.67	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-R1 (10 A)	10.00	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-Y2 (6 A)	16.67	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-B6 (10 A)	10.00	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-Y5 (16 A)	6.25	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-B7 (10 A)	10.00	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-R3 (40 A)	2.50	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-B3 (20 A)	5.00	10.7	0.8	Partial	**WARNING**

Upstream	Downstream	In ratio	Fault at downstream (kA)	Selectivity limit (kA)	Short circuit	Status
INC-DBGF1 (100 A)	DB-GF1-R6 (6 A)	16.67	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-B4 (16 A)	6.25	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-Y4 (16 A)	6.25	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-B5 (16 A)	6.25	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-R5 (10 A)	10.00	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-Y8 (6 A)	16.67	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-Y3 (20 A)	5.00	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-R4 (16 A)	6.25	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-B2 (6 A)	16.67	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-Y7 (6 A)	16.67	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-B1 (6 A)	16.67	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-R7 (6 A)	16.67	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-Y6 (16 A)	6.25	10.7	0.8	Partial	**WARNING**

DB SCHEDULE

Board	Cct	Circuit ID	Description	Load type	Connected (kW)	DF	Demand (kW)	PF	Ib (A)	Breaker	Poles	Icu (kA)	Cable	Length (m)	Vd total (%)	Status	Accessories
MDB-1	1	INC-GF	Incomer to Ground floor SMDB	Sub-board SMDB-GF					327	500 A MCCB	TP+N	36	4C × 300 mm² XLPE/SWA + 150 mm² CPC	35	0.46	PASS	
MDB-1	2	INC-FF	Incomer to First floor SMDB	Sub-board SMDB-FF					135	200 A MCCB	TP+N	36	4C × 95 mm² XLPE/SWA + 50 mm² CPC	55	0.75	PASS	
MDB-1	3	INC-MCC	Incomer to Pump room MCC	Sub-board MCC-1					210	315 A MCCB	TP+N	36	4C × 150 mm² XLPE/SWA + 95 mm² CPC	40	0.58	PASS	

Board	Cct	Circuit ID	Description	Load type	Connected (kW)	DF	Demand (kW)	PF	Ib (A)	Breaker	Poles	Icu (kA)	Cable	Length (m)	Vd total (%)	Status	Accessories
MDB-1	4	DB-EV	EV charging – parking	EV charging	44	0.7	30.8	0.98	44	63 A C	TP+N	36	4C × 10 mm² XLPE/SWA + 10 mm² CPC	60	2.38	PASS	
MDB-1	5	DB-PV	Solar PV inverter	Solar PV	50	1	50	1	70	100 A MCCB	TP+N	36	4C × 25 mm² XLPE/SWA + 16 mm² CPC	30	0.76	PASS	
MDB-1		**TOTAL**	**Main Distribution Board — 1600 A**		**514**		**451.1**	**0.88**	**712**			**29.2**				**45% loaded**	
SMDB-GF	1	GF-LTG	Lighting	Lighting	45	0.9	40.5	0.9	63	80 A MCCB	TP+N	25	4C × 16 mm² XLPE/SWA + 16 mm² CPC	30	1.47	PASS	
SMDB-GF	2	GF-SKT	Sockets	Small power / sockets	28	0.6	16.8	0.85	27	40 A C	TP+N	25	4C × 4 mm² XLPE/SWA + 4 mm² CPC	35	2.37	PASS	
SMDB-GF	3	GF-HVAC	HVAC	HVAC	120	0.9	108	0.85	177	250 A MCCB	TP+N	25	4C × 95 mm² XLPE/SWA + 50 mm² CPC	40	1.16	PASS	
SMDB-GF	4	INC-DBGF1	Incomer to Villa ground floor DB	Sub-board DB-GF1					60	100 A MCCB	TP+N	25	4C × 35 mm² XLPE/SWA + 16 mm² CPC	30	0.92	PASS	
SMDB-GF		**TOTAL**	**Ground floor SMDB — 630 A**		**231**		**203.3**	**0.87**	**325**			**21.8**				**52% loaded**	
DB-GF1	1	DB-GF1-Y1	Majlis lighting	Lighting	1	1	1	0.9	5	6 A C	SP+N	15	2C × 2.5 mm² XLPE/SWA + 2.5 mm² CPC	18	1.48	PASS	
DB-GF1	2	DB-GF1-R1	Living lighting	Lighting	1.2	1	1.2	0.9	6	10 A C	SP+N	15	2C × 2.5 mm² XLPE/SWA + 2.5 mm² CPC	20	1.67	PASS	

Board	Cct	Circuit ID	Description	Load type	Connected (kW)	DF	Demand (kW)	PF	Ib (A)	Breaker	Poles	Icu (kA)	Cable	Length (m)	Vd total (%)	Status	Accessories
DB-GF1	3	DB-GF1-Y2	Kitchen lighting	Lighting	0.8	1	0.8	0.9	4	6 A C	SP+N	15	2C × 2.5 mm² XLPE/SWA + 2.5 mm² CPC	22	1.49	PASS	
DB-GF1	4	DB-GF1-B6	Majlis sockets	Small power / sockets	1.5	1	1.5	0.9	7	10 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	18	1.44	PASS	
DB-GF1	5	DB-GF1-Y5	Living sockets	Small power / sockets	2	1	2	0.9	9	16 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	20	1.7	PASS	
DB-GF1	6	DB-GF1-B7	Kitchen sockets	Small power / sockets	1.3	1	1.3	0.9	6	10 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	22	1.45	PASS	
DB-GF1	7	DB-GF1-R3	Cooker	General	6	1	6	0.9	28	40 A C	SP+N	15	2C × 6 mm² XLPE/SWA + 6 mm² CPC	24	2.79	PASS	
DB-GF1	8	DB-GF1-B3	Kitchen water heater	General	3	1	3	0.9	14	20 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	24	2.32	PASS	
DB-GF1	9	DB-GF1-R6	Maid room	Small power / sockets	1.1	1	1.1	0.9	5	6 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	28	1.51	PASS	
DB-GF1	10	DB-GF1-B4	Majlis split A/C	HVAC	2.5	1	2.5	0.9	12	16 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	18	1.79	PASS	
DB-GF1	11	DB-GF1-Y4	Living split A/C	HVAC	2.5	1	2.5	0.9	12	16 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	20	1.89	PASS	
DB-GF1	12	DB-GF1-B5	Dining split A/C	HVAC	2.5	1	2.5	0.9	12	16 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	22	1.99	PASS	

Board	Cct	Circuit ID	Description	Load type	Connected (kW)	DF	Demand (kW)	PF	Ib (A)	Breaker	Poles	Icu (kA)	Cable	Length (m)	Vd total (%)	Status	Accessories
DB-GF1	13	DB-GF1-R5	Guest bedroom	Small power / sockets	1.5	1	1.5	0.9	7	10 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	26	1.67	PASS	
DB-GF1	14	DB-GF1-Y8	Guest bathroom	Lighting	0.3	1	0.3	0.9	1	6 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	27	1.05	PASS	
DB-GF1	15	DB-GF1-Y3	Guest bath water heater	General	3	1	3	0.9	14	20 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	27	2.49	PASS	
DB-GF1	16	DB-GF1-R4	Guest bedroom split A/C	HVAC	2.5	1	2.5	0.9	12	16 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	26	2.18	PASS	
DB-GF1	17	DB-GF1-B2	Dining lighting	Lighting	0.6	1	0.6	0.9	3	6 A C	SP+N	15	2C × 2.5 mm² XLPE/SWA + 2.5 mm² CPC	22	1.33	PASS	
DB-GF1	18	DB-GF1-Y7	Dining sockets	Small power / sockets	1	1	1	0.9	5	6 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	22	1.35	PASS	
DB-GF1	19	DB-GF1-B1	External lighting	Lighting	1	1	1	0.9	5	6 A C	SP+N	15	2C × 2.5 mm² XLPE/SWA + 2.5 mm² CPC	35	2.01	PASS	
DB-GF1	20	DB-GF1-R7	Water pump	Motor	0.8	1	0.8	0.9	3	6 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	30	1.36	PASS	
DB-GF1	21	DB-GF1-Y6	Laundry	Small power / sockets	2	1	2	0.9	9	16 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	25	1.89	PASS	
DB-GF1		**TOTAL**	**Villa ground floor DB — 100 A**		**38**		**38**	**0.9**	**59**			**10.7**				**59% loaded**	

Board	Cct	Circuit ID	Description	Load type	Connected (kW)	DF	Demand (kW)	PF	Ib (A)	Breaker	Poles	Icu (kA)	Cable	Length (m)	Vd total (%)	Status	Accessories
SMDB-FF	1	FF-OFF	Office	Small power / sockets	35	0.8	28	0.9	43	63 A C	TP+N	25	4C × 10 mm² XLPE/SWA + 10 mm² CPC	25	1.66	PASS	
SMDB-FF	2	FF-IT	IT room	IT / data	20	1	20	0.95	29	40 A C	TP+N	25	4C × 4 mm² XLPE/SWA + 4 mm² CPC	20	2.04	PASS	
SMDB-FF	3	FF-LAB	Lab	General	50	0.7	35	0.85	57	80 A MCCB	TP+N	25	4C × 16 mm² XLPE/SWA + 16 mm² CPC	30	1.62	PASS	
SMDB-FF	4	FF-LTG	Corridor lighting	Lighting	4	1	4	0.9	19	25 A C	SP+N	25	2C × 6 mm² XLPE/SWA + 6 mm² CPC	35	2.57	PASS	
SMDB-FF		**TOTAL**	**First floor SMDB — 250 A**		**109**		**87**	**0.89**	**135**			**14.3**				**54% loaded**	
MCC-1	1	MCC-WP	Water pump	Motor	75	1	75	0.86	121	160 A MCCB	TP+N	25	4C × 50 mm² XLPE/SWA + 25 mm² CPC	20	1.03	PASS	
MCC-1	2	MCC-FP	Fire pump	Fire pump	55	1	55	0.86	89	125 A MCCB	TP+N	25	4C × 35 mm² FR + 16 mm² CPC	25	1.12	PASS	
MCC-1		**TOTAL**	**Pump room MCC — 400 A**		**130**		**130**	**0.86**	**210**			**19**				**53% loaded**	

CABLE SCHEDULE

Cable tag	From	To	Ref	Type	Cores × size	CPC (mm²)	Length (m)	Tray route	Ib (A)	Breaker In (A)	Iz (A)	Vd cable (%)	Vd total (%)	Status
C-INC-GF	MDB-1	SMDB-GF	1	Cu/XLPE/SWA/PVC (armoured)	4C × 300 mm²	150	35		327	500	525	0.46	0.46	PASS
C-INC-FF	MDB-1	SMDB-FF	6	Cu/XLPE/SWA/PVC (armoured)	4C × 95 mm²	50	55		135	200	266	0.75	0.75	PASS

Cable tag	From	To	Ref	Type	Cores × size	CPC (mm²)	Length (m)	Tray route	Ib (A)	Breaker In (A)	Iz (A)	Vd cable (%)	Vd total (%)	Status
C-INC-MCC	MDB-1	MCC-1	4	Cu/XLPE/SWA/PVC (armoured)	4C × 150 mm²	95	40		210	315	342	0.58	0.58	PASS
C-DB-EV	MDB-1	EV charging – parking	12	Cu/XLPE/SWA/PVC (armoured)	4C × 10 mm²	10	60		44	63	70	2.38	2.38	PASS
C-DB-PV	MDB-1	Solar PV inverter	10	Cu/XLPE/SWA/PVC (armoured)	4C × 25 mm²	16	30		70	100	120	0.76	0.76	PASS
C-GF-LTG	SMDB-GF	Lighting	11	Cu/XLPE/SWA/PVC (armoured)	4C × 16 mm²	16	30		63	80	91	1	1.47	PASS
C-GF-SKT	SMDB-GF	Sockets	14	Cu/XLPE/SWA/PVC (armoured)	4C × 4 mm²	4	35		27	40	41	1.91	2.37	PASS
C-GF-HVAC	SMDB-GF	HVAC	6	Cu/XLPE/SWA/PVC (armoured)	4C × 95 mm²	50	40		177	250	266	0.7	1.16	PASS
C-INC-DBGF1	SMDB-GF	DB-GF1	9	Cu/XLPE/SWA/PVC (armoured)	4C × 35 mm²	16	30		60	100	146	0.45	0.92	PASS
C-DB-GF1-Y1	DB-GF1	Majlis lighting	28	Cu/XLPE/SWA/PVC (armoured)	2C × 2.5 mm²	2.5	18		5	6	31	0.56	1.48	PASS
C-DB-GF1-R1	DB-GF1	Living lighting	28	Cu/XLPE/SWA/PVC (armoured)	2C × 2.5 mm²	2.5	20		6	10	31	0.75	1.67	PASS
C-DB-GF1-Y2	DB-GF1	Kitchen lighting	28	Cu/XLPE/SWA/PVC (armoured)	2C × 2.5 mm²	2.5	22		4	6	31	0.58	1.49	PASS
C-DB-GF1-B6	DB-GF1	Majlis sockets	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	18		7	10	41	0.52	1.44	PASS
C-DB-GF1-Y5	DB-GF1	Living sockets	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	20		9	16	41	0.78	1.7	PASS
C-DB-GF1-B7	DB-GF1	Kitchen sockets	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	22		6	10	41	0.53	1.45	PASS
C-DB-GF1-R3	DB-GF1	Cooker	25	Cu/XLPE/SWA/PVC (armoured)	2C × 6 mm²	6	24		28	40	52	1.88	2.79	PASS
C-DB-GF1-B3	DB-GF1	Kitchen water heater	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	24		14	20	41	1.4	2.32	PASS

Cable tag	From	To	Ref	Type	Cores × size	CPC (mm²)	Length (m)	Tray route	Ib (A)	Breaker In (A)	Iz (A)	Vd cable (%)	Vd total (%)	Status
C-DB-GF1-R6	DB-GF1	Maid room	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	28		5	6	41	0.59	1.51	PASS
C-DB-GF1-B4	DB-GF1	Majlis split A/C	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	18		12	16	41	0.87	1.79	PASS
C-DB-GF1-Y4	DB-GF1	Living split A/C	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	20		12	16	41	0.97	1.89	PASS
C-DB-GF1-B5	DB-GF1	Dining split A/C	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	22		12	16	41	1.07	1.99	PASS
C-DB-GF1-R5	DB-GF1	Guest bedroom	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	26		7	10	41	0.75	1.67	PASS
C-DB-GF1-Y8	DB-GF1	Guest bathroom	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	27		1	6	41	0.14	1.05	PASS
C-DB-GF1-Y3	DB-GF1	Guest bath water heater	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	27		14	20	41	1.57	2.49	PASS
C-DB-GF1-R4	DB-GF1	Guest bedroom split A/C	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	26		12	16	41	1.26	2.18	PASS
C-DB-GF1-B2	DB-GF1	Dining lighting	28	Cu/XLPE/SWA/PVC (armoured)	2C × 2.5 mm²	2.5	22		3	6	31	0.41	1.33	PASS
C-DB-GF1-Y7	DB-GF1	Dining sockets	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	22		5	6	41	0.43	1.35	PASS
C-DB-GF1-B1	DB-GF1	External lighting	28	Cu/XLPE/SWA/PVC (armoured)	2C × 2.5 mm²	2.5	35		5	6	31	1.09	2.01	PASS
C-DB-GF1-R7	DB-GF1	Water pump	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	30		3	6	41	0.44	1.36	PASS
C-DB-GF1-Y6	DB-GF1	Laundry	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	25		9	16	41	0.97	1.89	PASS
C-FF-OFF	SMDB-FF	Office	12	Cu/XLPE/SWA/PVC (armoured)	4C × 10 mm²	10	25		43	63	70	0.91	1.66	PASS
C-FF-IT	SMDB-FF	IT room	14	Cu/XLPE/SWA/PVC (armoured)	4C × 4 mm²	4	20		29	40	41	1.29	2.04	PASS

Cable tag	From	To	Ref	Type	Cores × size	CPC (mm²)	Length (m)	Tray route	Ib (A)	Breaker In (A)	Iz (A)	Vd cable (%)	Vd total (%)	Status
C-FF-LAB	SMDB-FF	Lab	11	Cu/XLPE/SWA/PVC (armoured)	4C × 16 mm²	16	30		57	80	91	0.87	1.62	PASS
C-FF-LTG	SMDB-FF	Corridor lighting	25	Cu/XLPE/SWA/PVC (armoured)	2C × 6 mm²	6	35		19	25	52	1.82	2.57	PASS
C-MCC-WP	MCC-1	Water pump	8	Cu/XLPE/SWA/PVC (armoured)	4C × 50 mm²	25	20		121	160	174	0.45	1.03	PASS
C-MCC-FP	MCC-1	Fire pump	9 *	Fire-rated, BS 8491 (120 min, large cables)	4C × 35 mm²	16	25		89	125	146	0.54	1.12	PASS

ASSUMPTIONS AND LIMITATIONS

- Upstream MV network treated as infinite; transformer %Z split into R and X by its X/R ratio (default 5).
- Fault levels follow IEC 60909-0 in a simplified form, not a full study: maximum currents for breaking capacity with c_{max} = 1.05 and cable R at 20 °C; minimum currents for earth-fault disconnection with c_{min} = 0.95 and cable R at operating temperature; transformer correction K_T not applied; loads, motor contribution and generation are neglected.
- Cable ratings: the rating for each circuit's IEC 60364-5-52 installation method (E free air / tray unless set), derated for air ambient or, for buried cables, ground temperature and soil thermal resistivity, and for grouping (the cable tray route for method E, else the circuit's own parallel runs). A circuit without a rating for its method is DATA REQUIRED.
- Cable resistance at operating temperature = 1.2 × R20 (IEC 60228); protective conductor per IEC 60364-5-54 Table 54.2 unless specified. Maximum fault levels use R at 20 °C.
- Selectivity is assessed from current thresholds only; confirm with the breaker manufacturer's selectivity tables.
- Busbar voltages are measured from the main board busbar and exclude transformer regulation.

Generated by LV Design Studio 1.38.0. Calculated checks against the project limits — engineering review and approval are separate. Results must be checked by a qualified engineer before use. **Cable data — 2.5, 4, 6, 10, 16, 25, 35, 50, 95, 150, 300 mm²: app reference value (approximate, not manufacturer-verified). Verify against the manufacturer's datasheet before relying on these results.**

CALCULATION RECORD

What these results were calculated from, with what, and how far they can be relied on.

Project	Villa Complex 415V LV Network Project ID not assigned (save the project) · file format version 1
Calculated	At the time of this print, from the current inputs
Software	LV Design Studio 1.38.0 · built-in calculation engine (no external engine used)
Calculation basis	Methods per Codes and standards → Calculation basis and each study's Method notes. Standard editions: not verified against a specific edition
Key assumptions	• 415 V, 3-phase + N, 50 Hz, TN-S · design ambient 45 °C · voltage drop limit 4 % source to load • Cable resistance at operating temperature 1.2 × R20 (not set for the project)

	• Fault levels: maximum Ik″ with c max = 1.05 and cable R at 20 °C, MV network infinite, transformer correction KT not applied — **simplified, preliminary, not a full IEC 60909 study**
Cable data	**11 sizes used: 11 app reference values (approximate, not manufacturer-verified) · 0 manufacturer data** Final-design requirement for manufacturer cable data: not set
Input checks	No invalid inputs, no supply loops, no missing references
Calculation warnings	0 failures · 0 to check (see the results summary on page 1)
Limitations	• Cable ratings: per installation method with ambient / ground, soil and grouping factors; grouping with other circuits is counted only on cable tray routes • Selectivity from current thresholds only — confirm with the manufacturer's tables • Voltage drop: first-order, summed along the radial path; not an unbalanced load flow

Validation status: PRELIMINARY. Calculated checks against the project limits, using approximate reference or unverified cable data. Not an engineering approval and not an approval by DEWA or any other authority.